

Manuscript Title: A High-Resolution Finite-Difference Benchmark for Steady Continuation and Hopf Bifurcation Dynamics up to Extreme Reynolds Numbers ($Re = 100,000$) in 2D Lid-Driven Cavity Flows

Author: Kartikey Singh | Department of Mathematics, University of Delhi, Delhi, 110007, India
Email: kartikeysingh525@protonmail.com | ORCID: <https://orcid.org/0009-0009-1973-7532>

Dear Editor-in-Chief and Editorial Board,

We are pleased to submit our original research manuscript entitled “**A High-Resolution Finite-Difference Benchmark for Steady Continuation and Hopf Bifurcation Dynamics up to Extreme Reynolds Numbers ($Re = 100,000$) in 2D Lid-Driven Cavity Flows**” for consideration as a regular research article in *Computers & Fluids*.

Significance and Key Contributions:

The incompressible 2D lid-driven cavity flow remains the canonical proving ground for computational fluid dynamics. However, established benchmarks stagnate at $Re \leq 10,000$ (Ghia et al., 1982) or $Re \leq 25,000$ (Erturk et al., 2005). In this manuscript, we push the numerical and scientific frontiers through several key contributions:

1. **Exact Spectral Poisson Acceleration:** We integrate an exact Discrete Sine Transform (DST-I) Poisson solver operating at optimal $\mathcal{O}(N^2 \log N)$ complexity, eliminating iterative residual drift, pressure-velocity decoupling, and numerical dissipation.
2. **Asymptotic Core Vorticity Homogenization ($Re \rightarrow 100,000$):** We compute steady continuation solutions up to $Re = 100,000$ on ultra-fine grids (513×513) and a high-resolution grid (1025×1025 , 1,050,625 nodes). We verify Batchelor’s (1956) asymptotic recirculating core theorem through the plateau $\omega_0 = -2.5418$ alongside symmetric mid-plane alignment ($x_c \rightarrow 0.50000$).
3. **Localized Hybrid Differencing & Dissipation Mapping:** We incorporate a localized Spalding–Patankar hybrid differencing scheme at $Re = 100,000$ ($Pe = 195.3 \gg 2$) that regularizes corner singularity shocks while providing a rigorous 2D spatial audit of numerical viscosity ($\nu_{\text{num}}/\nu \leq 0.99$ in the core on 513×513 and 0.24 on 1025×1025), with pure central differencing ($\nu_{\text{num}} = 0$) strictly preserved everywhere for all $Re \leq 50,000$.
4. **Extended Unsteady Marching & Bifurcation Cascades:** Through time-accurate physical marching on 513×513 meshes ($t = 16.0$ – 22.0), we establish the primary supercritical Hopf limit cycle over extended convective time ($St = 0.6664$, recurrence error $\epsilon_{\text{rec}} < 0.08\%$), capture secondary Hopf 2-torus quasi-periodicity ($St = 0.5554$, 10.0 cycles), and resolve multiple complete periodic cycles across high Reynolds numbers: $Re = 25,000$ ($St = 0.1124$, 2.0 cycles), $Re = 50,000$ ($St = 0.2399$, 4.0 cycles), and extreme $Re = 100,000$ ($St = 0.2272$, 3.0 cycles).
5. **Scientific Reproducibility & Open Benchmarks:** We openly release 28 full 2D coordinate/flow field datasets, phase portrait scripts, and automated visualization tools under the MIT License with permanent Zenodo DOIs.

Suggested Reviewers:

1. **Prof. Ercan Erturk** (Kocaeli University / ITU, Turkey) — erturke@itu.edu.tr
2. **Prof. Olivier Botella** (Université de Lorraine, LEMTA, France) — olivier.botella@univ-lorraine.fr
3. **Prof. Chang Shu** (National University of Singapore, Singapore) — mpeshuc@nus.edu.sg
4. **Prof. Charles-Henri Bruneau** (Université de Bordeaux, France) — charles-henri.bruneau@u-bordeaux.fr

Declarations: This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All data and tools are openly available. There are no competing financial or non-financial interests to declare.

Sincerely yours,

Kartikey Singh

Department of Mathematics, University of Delhi, Delhi, 110007, India
Email: kartikeysingh525@protonmail.com